# MONOMORPHISMS OF ALGEBRAIC STACKS ARE INJECTIVE ON POINTS

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1732-1739 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 1741-1754, the complete author proof (12 body lines).

*Editorial note.* Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 25 of queue_stacks-properties.json (pb=662, stmt 1732-1739, proof 1741-1754).

*Editorial note.* stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

*Editorial note.* Preamble patch operations/agent-authoring/lane-s3/patch-sd-envs.tex appended (sd02 precedent): the author statement carries a slogan environment with real content, which the comment-package scan cannot swallow; the patch turns the envs into empty containers.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma lemma-monomorphism: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1674–1687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Monomorphisms of stacks are injective on points. A monomorphism of algebraic stacks induces an injective map of sets of points.*

*Proof.* Let $f : \mathcal{X} \to \mathcal{Y}$ be a monomorphism of algebraic stacks. Suppose that $x_i : \mathrm{Spec}(K_i) \to \mathcal{X}$ be morphisms such that $f \circ x_1$ and $f \circ x_2$ define the same element of $|\mathcal{Y}|$. Applying the definition we find a common extension $\Omega$ with corresponding morphisms $c_i : \mathrm{Spec}(\Omega) \to \mathrm{Spec}(K_i)$ and a 2-isomorphism $\beta : f \circ x_1 \circ c_1 \to f \circ x_1 \circ c_2$. As $f$ is fully faithful, see Lemma lemma-monomorphism, we can lift $\beta$ to an isomorphism $\alpha : x_1 \circ c_1 \to x_1 \circ c_2$. Hence $x_1$ and $x_2$ define the same point of $|\mathcal{X}|$ as desired. □

*Date*: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.